



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	The company is responsible for providing services such as web design, graphic design, and social media marketing solutions for small businesses. While everyone was doing their work, the company's services suddenly stopped operating. To solve this problem, the company's cybersecurity team was called. After investigating, they discovered that the company's server responsible for these services had suffered a hacker attack. The process of stopping the attack took two hours. First, the team investigated the type of attack performed so they could implement security measures to contain the damage.
Identify	The hacker targeted the organization's server, which is responsible for managing and storing all the company's service information. The attacker performed a type of Denial of Service (DoS) attack called an Internet Control Message Protocol (ICMP) flood attack, where the target device is flooded with ICMP requests. ICMP is a protocol used to send data packets and communicate transmission errors across a network. This overload of ICMP requests caused the organization's server and services to stop operating and crash.
Protect	The immediate plan used by the cybersecurity team to stop the attack was carried out in two steps. The first step was limiting the rate of incoming ICMP packets, which prevented the server from receiving the excessive number of

	packets that caused the crash. The second step was dealing with the IP spoofing technique, where the cybersecurity team configured the firewall and blocked the IP source responsible for sending the ICMP packets.
Detect	To detect a possible attack, the security team will need to rely on two tools that work together to detect malicious activity at any moment. The first one is an Intrusion Detection System (IDS), which is a tool that automatically detects and alerts possible intrusions. The second one is a Security Information and Event Management (SIEM) system, which is used by cybersecurity professionals to collect and analyze log data in real time from multiple network machines. By using these tools, the cybersecurity team will be able to identify an attack before it causes serious damage.
Respond	To respond after an incident occurs, the first process can be done using a tool called packet sniffing. Once the DoS attack has been performed, the cybersecurity team can analyze the header section of the packets causing the incident and identify the hacker's IP source. With the IP source in hand, the team can use the firewall to block any requests coming from that specific IP address, containing the attack immediately. If blocking the source is not enough, the firewall can also be configured to limit ICMP requests.
Recover	After the incident, the team must restore the company's server to normal operation and ensure that the services are working properly. They should also check whether the hacker performed another attack during the DoS attack, such as a data breach or malware deployment. If so, it must be contained and removed as soon as possible.
